

MERN E-Commerce Backend - Production Ready Project

Complete Architecture & Full REST API Implementation Blueprint

Designed for Next.js 15 App Router Frontend

Tech Stack: Node.js, Express.js, MongoDB, Mongoose, JWT, Multer, Cloudinary, Razorpay, Nodemailer

Project Root: C:\Users\User\.gemini\antigravity\scratch\ecommerce-backend

Complete Backend Modules Included in Codebase:

1. Base API & Server Setup: Express app, CORS, Helmet, Rate Limiter, Mongoose db.js, ApiError, ApiResponse
2. Auth & User Management: User model, Bcrypt password hashing, JWT Access & Refresh tokens, Cookies, Roles
3. Category & Product Catalog: Category model, Product model, Filtering, Search, Pagination, Ratings & Reviews
4. Cart & Order Management: Multi-item Cart model, Checkout, Stock deduction, Status management, Admin orders
5. Payment Gateway: Razorpay Checkout Order creation & HMAC SHA256 signature verification
6. Media Uploads: Multer file parser with Cloudinary upload/delete helper functions
7. Email Notifications: Nodemailer SMTP wrapper with HTML email templates for orders & password reset

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